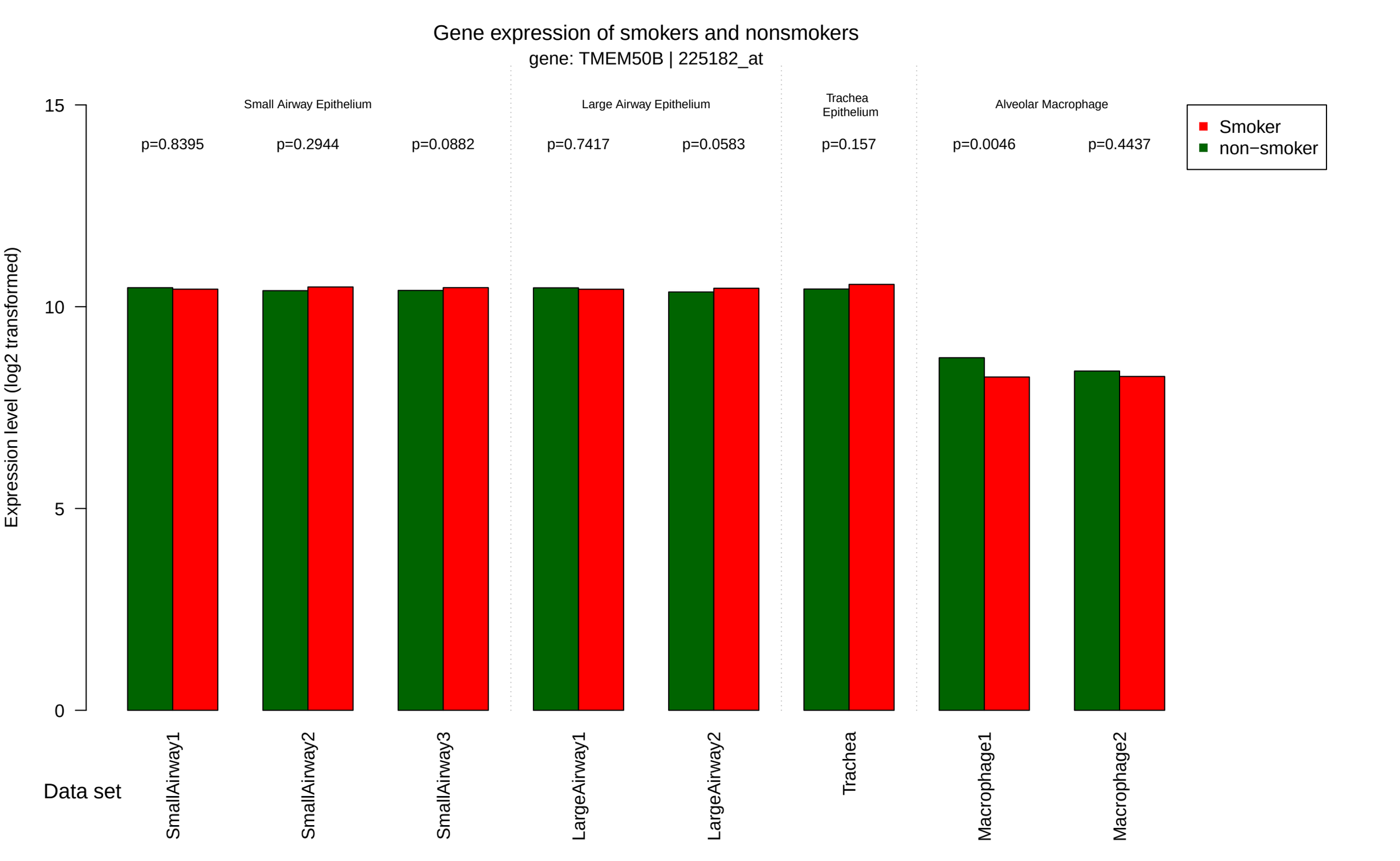
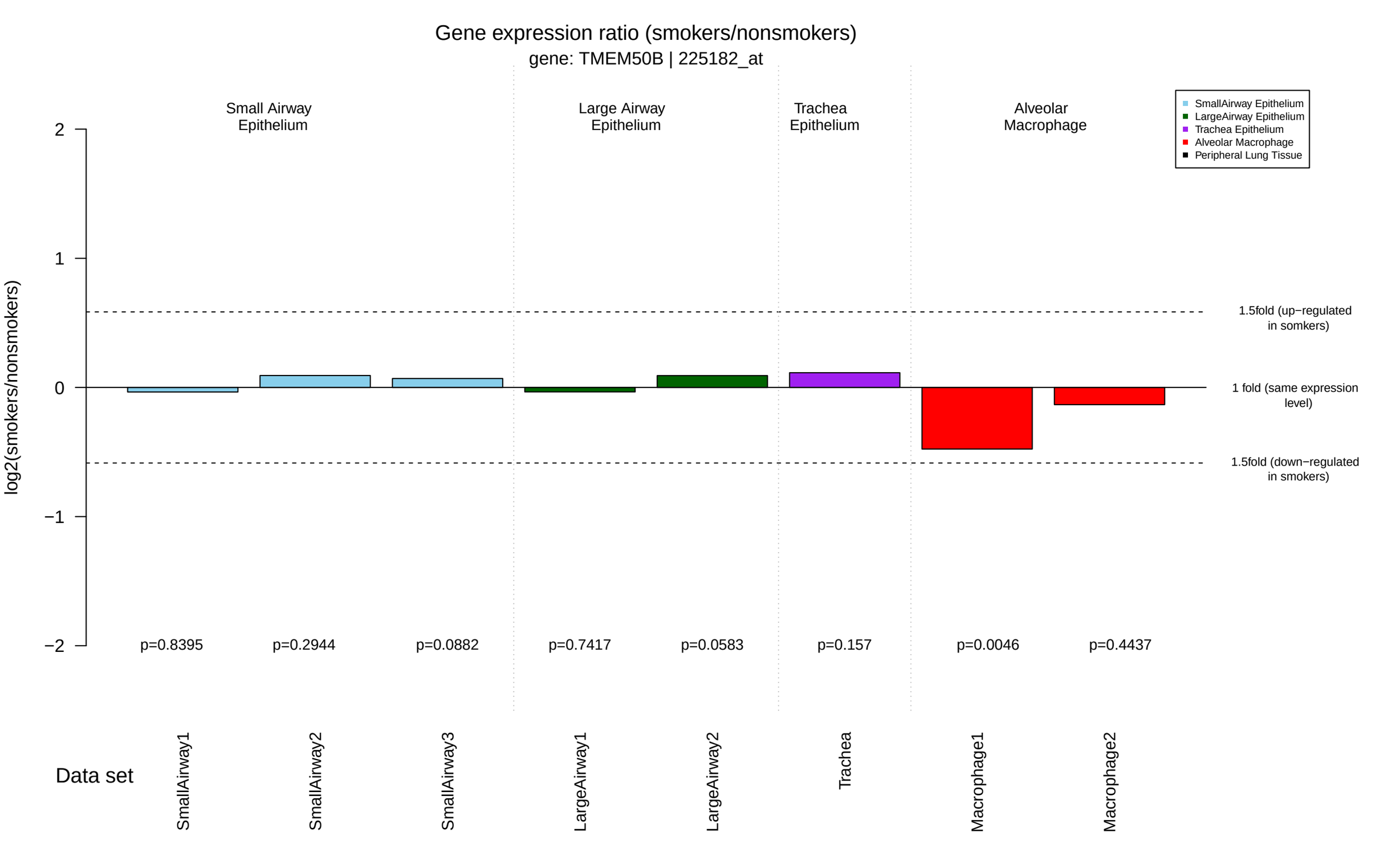
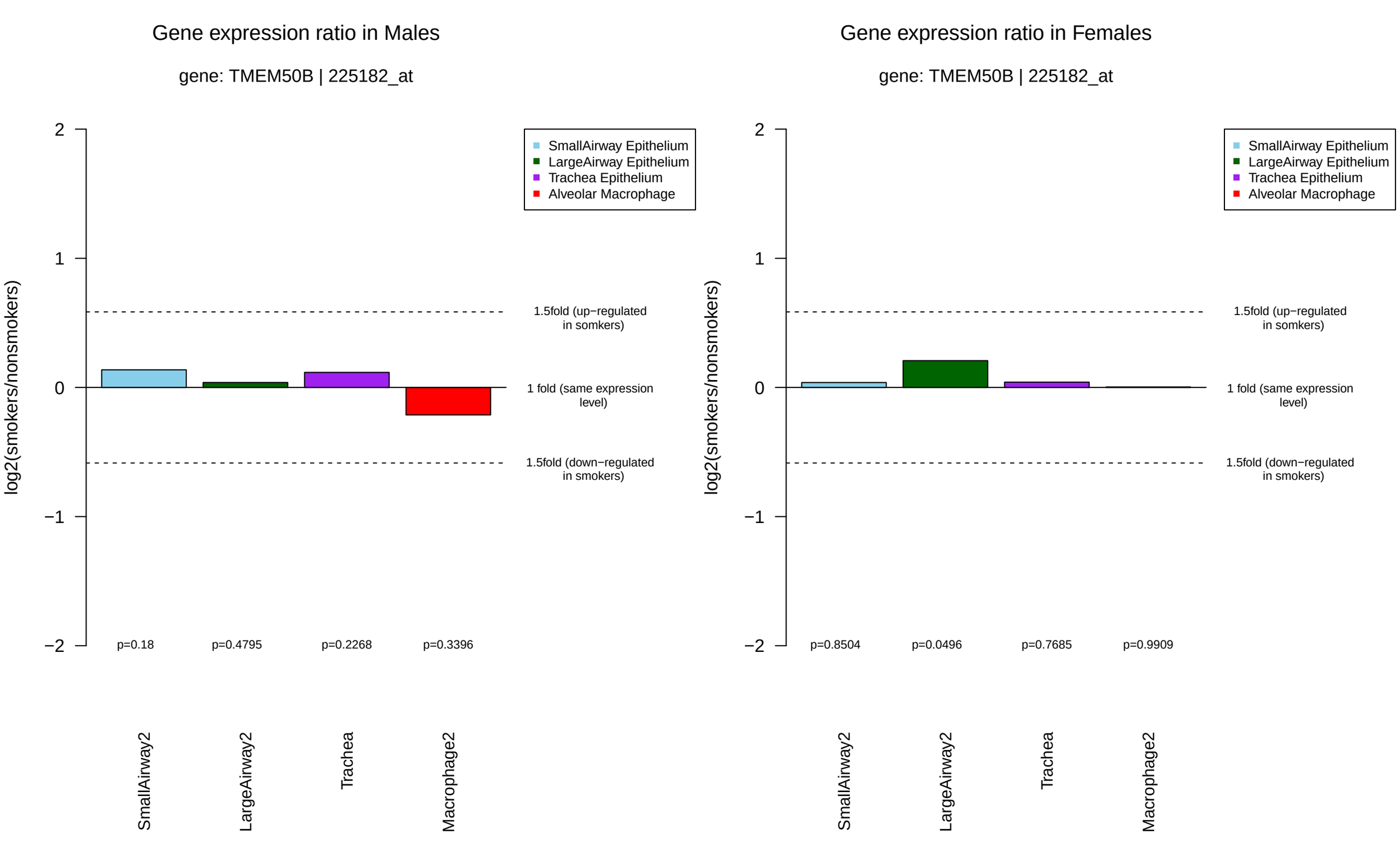
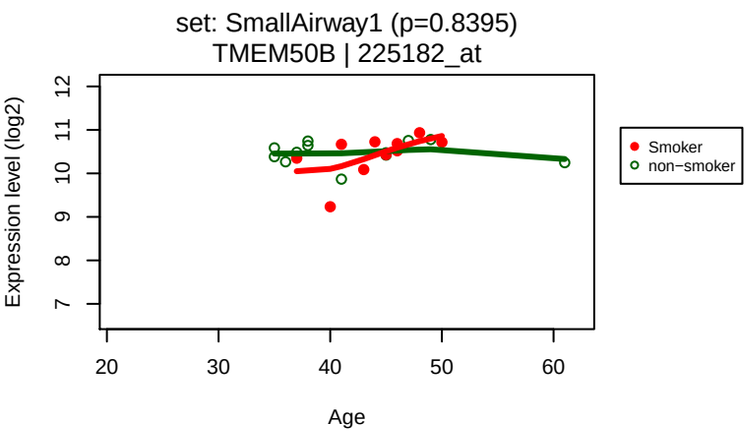
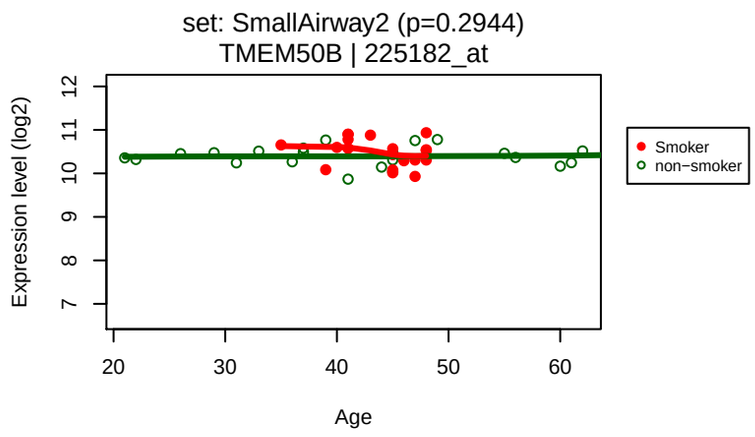
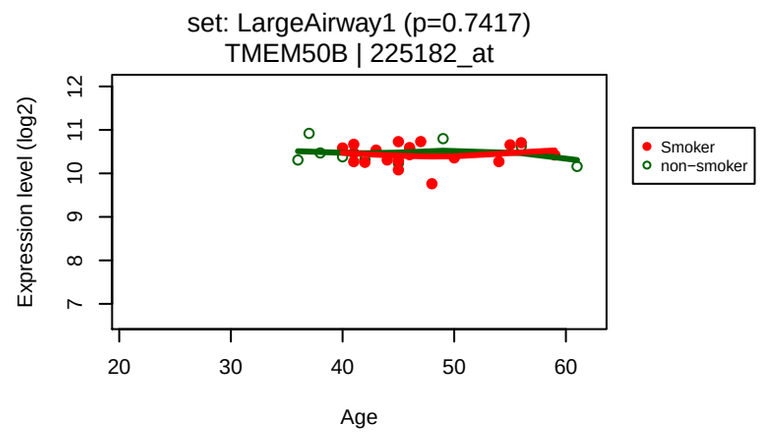

Gene expression of smokers and nonsmokers
gene: TMEM50B | 225182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8395
p=0.2944
p=0.0882
p=0.7417
p=0.0583
p=0.157
p=0.0046
p=0.4437
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

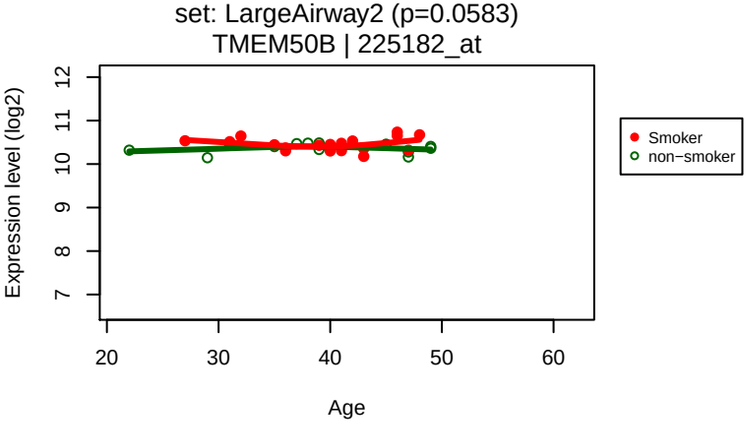

Gene expression ratio (smokers/nonsmokers)
gene: TMEM50B | 225182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8395
p=0.2944
p=0.0882
p=0.7417
p=0.0583
p=0.157
p=0.0046
p=0.4437
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

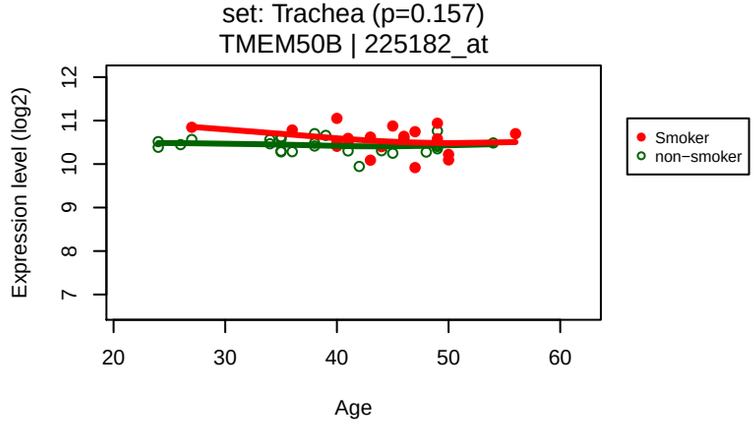

Gene expression ratio in Males
gene: TMEM50B | 225182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.18
p=0.4795
p=0.2268
p=0.3396
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM50B | 225182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8504
p=0.0496
p=0.7685
p=0.9909
SmallAirway2
LargeAirway2
Trachea
Macrophage2

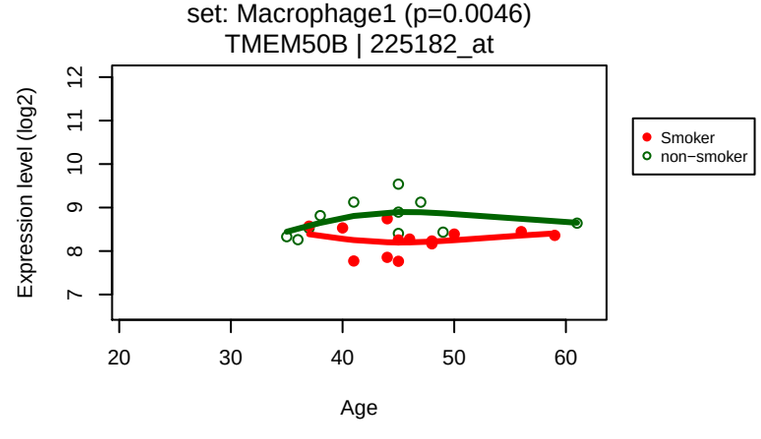

set: SmallAirway1 (p=0.8395)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker

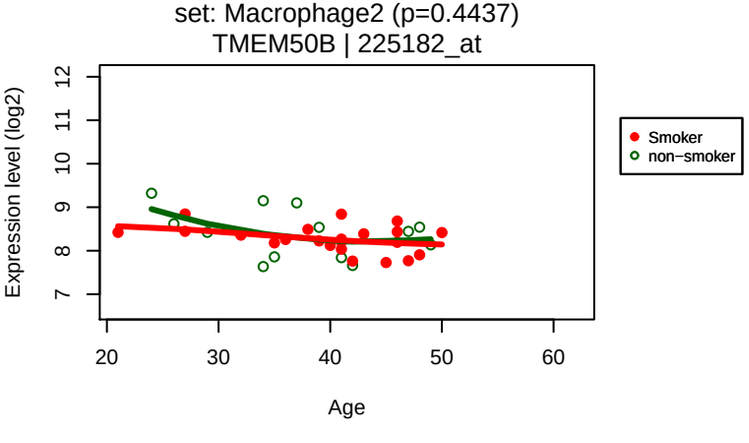

set: SmallAirway2 (p=0.2944)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker

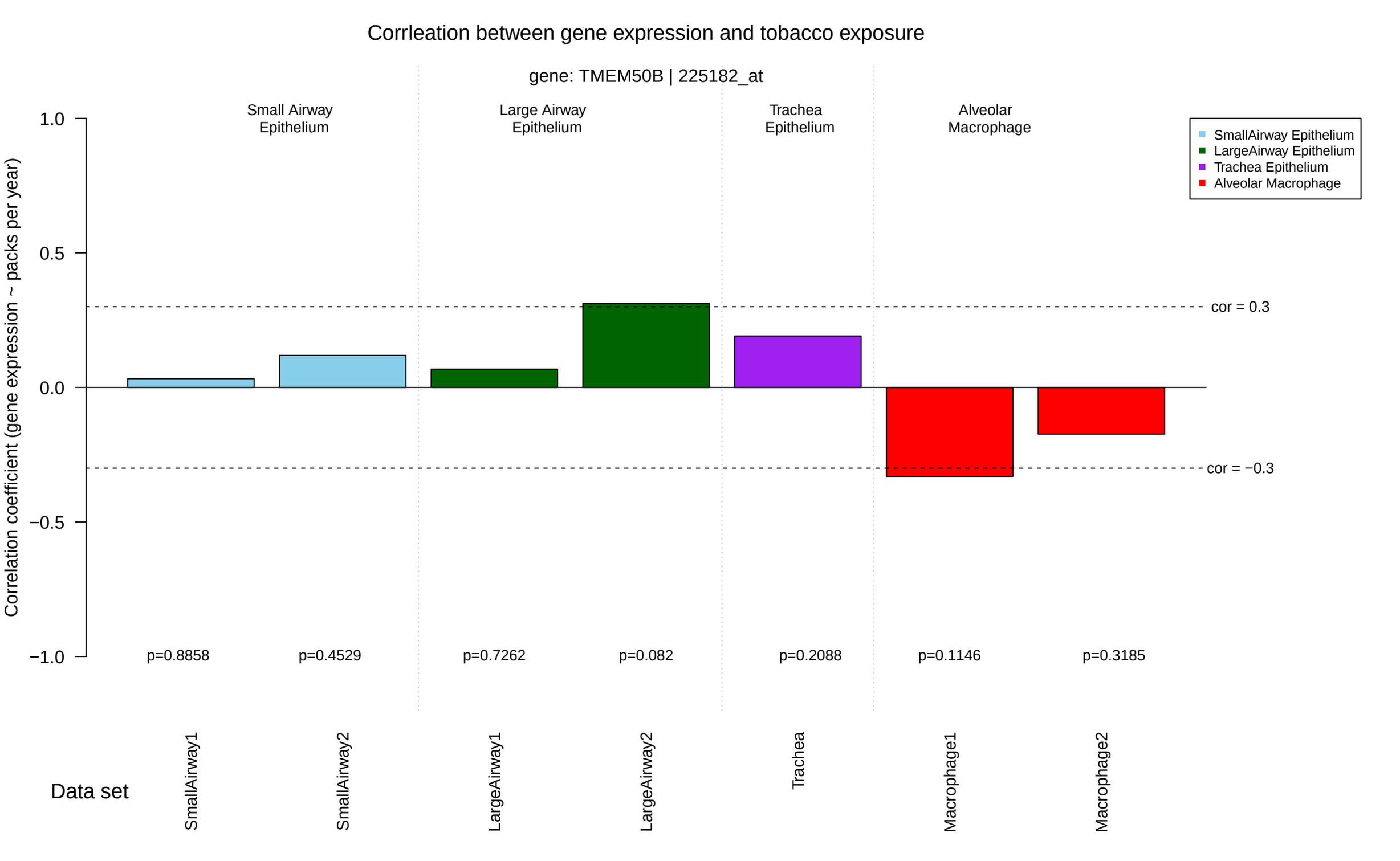

set: LargeAirway1 (p=0.7417)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0583)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.157)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0046)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4437)
TMEM50B | 225182_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM50B | 225182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8858
p=0.4529
p=0.7262
p=0.082
p=0.2088
p=0.1146
p=0.3185
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2